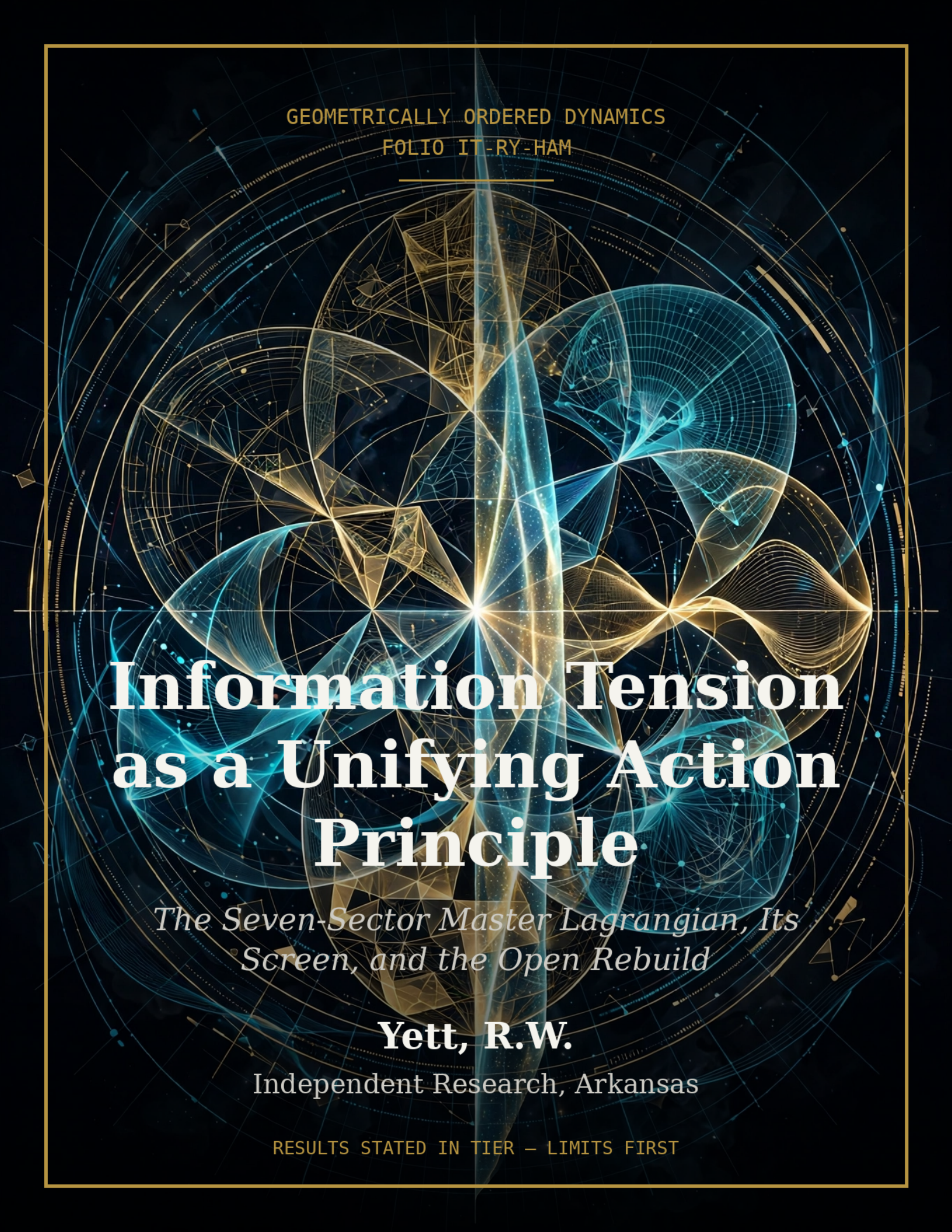
GEOMETRICALLY ORDERED DYNAMICS
FOLIO IT-RY-HAM



Information Tension as a Unifying Action Principle

The Seven-Sector Master Lagrangian, Its Screen, and the Open Rebuild

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RESULTS STATED IN TIER — LIMITS FIRST

CONVERGENCE -- INFORMATION TENSION

This folio converges on **information tension** as the unifying dynamical object of the program: the claim that one scalar field χ, read as the local gradient of geometric entropy, can carry gravitation, galaxy kinematics, gauge structure and the screen through a single action. Its unique angle is **the master action**: this paper states the seven-sector Lagrangian density in full, its equations of motion, and its screening limits, and marks every open seam in tier. Sibling successors read the same claim from other angles: [TIER: CONJECTURE] successor (A) reads it as an empirical law ($\mu_{\rm std}$ on SPARC), successor (G) reads it through layered architecture, and the Res Nova rebuild stream (Grok) is deriving the information-tension theory that this paper's PENDING sections wait on. No successor in this batch honestly represents **the Shadow** as a convergence target; that seat is open. Limits first: the substrate frame $V_{240}(\mathbb{R}^{57600})$ named in the source Lagrangian is retired by the source's own banner; nothing here claims the migrated derivation.

SUPERSESSION NOTE

Per the canon rule (one correct paper, one canon; a correction *becomes* the paper), this manuscript **supersedes Zenodo record `10.5281/zenodo.20652241`** ("The Yett Paradigm Complete", three-page record *On the Geometric Phase Force and the Spectral Gap of the Sovereign Manifold*). The old record's Sovereign Action, its χ-field definitions, and its mass-gap statement are carried forward below with their tiers stated; its overreaching claims (zero free parameters, machine-verified badges) are withdrawn here, not restated.

1 What this paper is

This is a **fixes-only synthesis**. No new claims, no new numbers, no new physics: every equation below is carried from one of three named sources, and every seam is labeled with the tier it actually holds. The three sources are:

1. The superseded record 20652241 [TIER: EMPIRICAL]—its Sovereign Action, its χ-field, and its mass-gap statement.
2. The Fully Expanded Lagrangian specification (BLK-986 child page, "Fully_Expanded_Lagrangian.tex") [TIER: CONJECTURE]—the seven-sector master density, the non-local kernel, the boundary sector, the equations of motion, and the screening and wave-speed limits. That document carries its own SUPERSEDED banner (2026-08-25), which is reproduced faithfully in §3.
3. The methodology pair, *referential-calculus.md* and *algorithm-as-poetry.md*, plus folio 26 (Lean equality as algebraic certificate) [TIER: CONJECTURE]—the method by which the paper is read, not theorems inside it.

What Grok's Res Nova rebuild stream has delivered to the Block as of this draft: **nothing yet**—the stream was assigned 2026-11-01 and its information-tension derivation has not landed. Sections depending on it are marked **PENDING** and are not filled by invention.

2 The information-tension field

Record 20652241 extends Einstein–Hilbert relativity with the Sovereign Information Tension field χ:

$$S_Y = \int_{\mathcal{M}} d^4x\,\sqrt{-g}\left[\frac{R}{16\pi G} - \frac{1}{2}g^{\mu\nu}\nabla_\mu\chi\nabla_\nu\chi - V(\chi) + 1 + f(\chi)\mathcal{L}_m\right], \tag{1}$$

with the geometric threshold $\chi_0 = 1/\sqrt{2}$ and

$$V(\chi) = \alpha(\chi-\chi_0)^2, \qquad f(\chi) = \beta(\chi-\chi_0)^2. \tag{2}$$

Variation with respect to the metric gives the modified field equations

$$G_{\mu\nu} = 8\pi G\,(1+f)\,T_{\mu\nu} + \nabla_\mu\chi\nabla_\nu\chi - g_{\mu\nu}\left[\tfrac{1}{2}(\nabla\chi)^2 + V(\chi)\right]. \tag{3}$$

The old record states that in the low-acceleration regime ($m_\chi r \ll 1$), spatial derivatives dominate and flat rotation curves follow without non-baryonic dark matter [TIER: CONJECTURE].

The Fully Expanded Lagrangian refines the kinematics. Defining the dimensionless kinetic gradient scalar

$$y(x) \equiv \frac{g^{\mu\nu}\nabla_\mu\chi\nabla_\nu\chi}{a_0^2/c^4} = \frac{\|\nabla\chi\|^2}{a_0^2}, \qquad a_0 = \frac{cH_0}{2\pi}, \tag{4}$$

the information-tension Lagrangian density is the AQUAL-type functional

$$\mathcal{L}_{\rm IT} = -\frac{c^4a_0^2}{8\pi G}\left[\sqrt{y(1+y)} - \operatorname{asinh}\sqrt{y}\right]. \tag{5}$$

Functional variation in y produces the interpolating function

$$\frac{\partial\mathcal{L}_{\rm IT}}{\partial y} = -\frac{c^4a_0^2}{8\pi G}\sqrt{\frac{y}{1+y}} \quad\xrightarrow{y=x^2}\quad \mu(x) = \frac{x}{\sqrt{1+x^2}}, \tag{6}$$

the $\mu_{\rm std}$ law whose empirical standing is the subject of successor (A), not of this paper. Photons and metric perturbations propagate on the disformal metric $\tilde{g}_{\mu\nu} = g_{\mu\nu} + 2\ell_P^2\,\nabla_\mu\chi\nabla_\nu\chi$.

3 The seven-sector master Lagrangian

The source Lagrangian states the total invariant action as a functional over spacetime $\mathcal{M}$ with boundary $\partial\mathcal{M}$:

$$\mathcal{S}_{\rm total} = \int_{\mathcal{M}} d^4x\,\sqrt{-g}\,\mathcal{L}_{\rm total} + \mathcal{S}_{\rm NL} + \mathcal{S}_{\rm boundary}, \tag{7}$$

where the density decomposes into seven coupled sectors:

$$\mathcal{L}_{\rm total} = \mathcal{L}_{\rm grav} + \mathcal{L}_{\rm dilaton} + \mathcal{L}_{\rm IT} + \mathcal{L}_{\rm Stiefel} + \mathcal{L}_{\rm gauge} + \mathcal{L}_{\rm fermion} + \mathcal{L}_{\rm Yukawa}. \tag{8}$$

1. **Conformal gravitation:** $\mathcal{L}_{\rm grav} = \dfrac{c^4}{16\pi G}e^{-2\phi}\,(R - 2\Lambda_0)$, a Weyl-frame coupling of the metric to the dilaton ϕ.
2. **Dilaton:** $\mathcal{L}_{\rm dilaton} = -\frac{1}{2}g^{\mu\nu}\nabla_\mu\phi\nabla_\nu\phi - \frac{1}{2}m_\phi^2\phi^2 - \frac{\lambda_\phi}{4!}\phi^4$.

3. **Information tension:** $\mathcal{L}_{\rm IT}$ as in §2—the paper's convergence center.
4. **Stiefel substrate:** kinetic term $\frac{1}{2}\,\mathrm{Tr}(g^{\mu\nu}\mathcal{D}_\mu V^\dagger \mathcal{D}_\nu V)$ with $\mathcal{D}_\mu = \nabla_\mu - iA_\mu$, minus the non-linear holonomy curvature trace coupling $\frac{\gamma(\chi)}{4}\,\mathrm{Tr}([V^\dagger\mathcal{D}_\mu V, V^\dagger \mathcal{D}_\nu V]^2)$, where

$$\gamma(\chi) = \gamma_0\,\frac{1-\chi}{1+\chi}, \qquad \lim_{\chi\to 0}\gamma(\chi) = \gamma_0 \text{ (GR limit)}, \qquad \lim_{\chi\to 1}\gamma(\chi) = 0 \text{ (dark-energy limit)}. \tag{9}$$

   This sector is stated on the retired substrate frame; see §3.
5. **Unified E_8 gauge:** $\mathcal{L}_{\rm gauge} = -\dfrac{1}{4g^2(\mu)}\,\mathrm{Tr}(F_{\mu\nu}F^{\mu\nu}) + \dfrac{\theta_{\rm QCD}}{32\pi^2}\,\mathrm{Tr}(F_{\mu\nu}\tilde{F}^{\mu\nu})$, with the source's stated bare Planck-scale matching

$$g_{\rm bare}(M_{\rm Pl}) = \frac{\kappa_Y\,\chi_Y}{\pi} = \frac{0.9539\times(1/\sqrt{2})}{\pi} \approx 0.2147. \tag{10}$$

   [TIER: CONJECTURE]
6. **Chiral fermions:** Dirac/Weyl spinors with nodal-harmonic collar mass matrix $M_i(\phi,\chi) = M_{\rm Pl}\,e^{\phi}\exp(-\lambda_i/\kappa_Y)\,[1+\xi_i(\chi-\chi_Y)]$, $\chi_Y = 1/\sqrt{2}$ [TIER: CONJECTURE].
7. **Yukawa:** $\sum_{f,f'} Y_{ff'}\,\phi\,\bar{\psi}_f\psi_{f'}$.

Two further pieces complete the action. The bi-local non-local tension kernel bridges coarse-grained surface density with point-wise stress-energy:

$$\mathcal{S}_{\rm NL} = \alpha\iint d^4x\,d^4y\,\sqrt{-g(x)}\sqrt{-g(y)}\;K(x,y)\,\frac{\rho_b(x)}{\Sigma_c}\,\frac{\rho_b(y)}{\Sigma_c}, \tag{11}$$

$K(x,y) = e^{-|x-y|/L_0}/(4\pi|x-y|)$. The boundary sector is Gibbons–Hawking–York plus an entanglement horizon term in the boundary quantum mutual information $I(A{:}B)$.

The substrate-frame banner

The source document's own SUPERSEDED banner (2026-08-25) is carried here verbatim in substance: the document names the big-dimension frame $V_{240}(\mathbb{R}^{57600})$ (or the 58,000 implementation variant) as its substrate; **that frame is retired**, and the arithmetic $240^2 = 57{,}600$ remains true only as an E_8 root-count identity, not as the ambient substrate dimension. No migrated derivation is claimed anywhere in this paper. **PENDING on Grok's stream:** the live-substrate Lagrangian.

4 Equations of motion and the screen

Variation in the inverse metric gives the holonomy-corrected Einstein equations,

$$G_{\mu\nu} + \Lambda_0 g_{\mu\nu} = \frac{8\pi G}{c^4}e^{2\phi}\left(T^{(\rm matter)}_{\mu\nu} + T^{(\phi)}_{\mu\nu} + T^{(\chi)}_{\mu\nu} + T^{(\rm Stiefel)}_{\mu\nu} + T^{(\rm gauge)}_{\mu\nu}\right), \tag{12}$$

with

$$T^{(\chi)}_{\mu\nu} = \frac{c^4}{8\pi G}\,\mu\!\left(\frac{\|\nabla\chi\|}{a_0}\right)\frac{\nabla_\mu\chi\nabla_\nu\chi}{a_0} - g_{\mu\nu}\mathcal{L}_{\rm IT}. \tag{13}$$

Variation in χ gives the non-linear constitutive equation

$$\nabla_\mu\left[\mu\!\left(\frac{\|\nabla\chi\|}{a_0}\right)\nabla^\mu\chi\right] = 4\pi G\,\rho_b + \frac{\gamma_0}{(1+\chi)^2}\,\mathcal{J}_{\rm Stiefel} + \xi_f\,\bar{\psi}\psi, \tag{14}$$

which the source states reproduces Poisson's equation at high accelerations and the Bekenstein–Milgrom field equation at low accelerations [TIER: CONJECTURE]. The stated screening limit is $\nabla\chi \to 0$; the stated tensor-speed bound is

$$c_T^2 = c^2\left(1 + \mathcal{O}(\ell_P^2\|\nabla\chi\|^2)\right) \implies \left|\frac{v_{\rm gw}}{c} - 1\right| < 10^{-15}, \tag{15}$$

compared by the source to GW170817/GRB 170817A [TIER: CONJECTURE].

5 Flagged tensions—tier-labeled, unresolved

(a) Spectral action vs. sector sum. Folio 45b states the unifying object is a **spectral action**, not a sum of Lagrangians. §3 above is a sum of seven Lagrangians. These two claims are in direct tension and this paper **does not reconcile them**: the seven-sector form is carried because it is the text the source supplies in full; the spectral-action claim is recorded as a live open question. **PENDING on Grok's stream:** exhibit the map or prove the non-isomorphism.

(b) 57,600 vs. 58,000. The source Lagrangian uses $V_{240}(\mathbb{R}^{57600})$ ($= 240^2$, self-consistent); folio 43 writes $V_{240}(\mathbb{R}^{58000})$; folio 45a and the conformal records use $N = 58{,}000$. RY ruled 2026-11-06: $N = 57{,}600$ is the real value (the 240^2 E_8 root-count identity); the 58,000 implementation count is superseded. Both frames are retired by the source banner in any case (§3).

(c) Single E_8 vs. $E_8 \times E_8$. The program's standing definition is $E_8 \times E_8$ heterotic, visible plus shadow sectors, the Shadow as the boundary. The source gauge sector was stated as a single E_8 ($\dim \mathfrak{e}_8 = 248 = 240 + 8$); by the Architect's order of 2026-11-02 it is re-typed here to $E_8 \times E_8$, the 240 roots realized on the visible-sector factor. The *landing* of the gauge sector in $E_8 \times E_8$ — the exhibited derivation, not the typing — is not given; it is recorded as open, not forced. **PENDING on the (G)-angle stream and Grok's rebuild.**

(d) The Yang–Mills gap statement as [conj]. Record 20652241 ties the gap to a hermitianized Berry–Keating operator and the Ramanujan tau function:

$$\Delta_{\rm YM} = \Delta_{\rm ERY} = \min(|\tau(2) - \tau(1)|,\, \alpha) > 0. \tag{16}$$

No map from the tau-function data into the physics framework is exhibited in the sources; per the isomorphism rule this stands as [TIER: CONJECTURE], proof open—not as a proved gap.

6 Method: how this paper is read

Two methodology inputs govern the reading, both tier-tagged as **method**, not as theorems about the physics.

Referential calculus. Every explored instance is read as the same $\mu F/\nu F$ skeleton: a finite reference, a lossy projection, an infinite referent, a conjugate-pair tradeoff, a certified invariant, and a bridge that makes convergence honest. The χ-field is read here as the finite reference whose lossy projection is the observed kinematics; the *bridge*—the theorem that would make convergence honest—is exactly what the PENDING sections await.

Algorithms as poetry. The initial-algebra/final-coalgebra formalization (algorithm $= (\mu F, \nu F,$ exec, $\langle\langle\cdot\rangle\rangle,$ fix)) is the method's formal idiom: write the finite blueprint, unfold the dynamics, certify the fixed point. This paper's blueprint is the seven-sector density; the dynamics are its equations of motion; the fixed point—the screen—is stated, not certified.

The certificate rule (folio 26). A Lean equality is an algebraic certificate, not a physical derivation. Accordingly this paper carries no "machine-verified" badge: the old record's verification claims are withdrawn, and the Lean proof roots named in the source Lagrangian (`ITT_Definitions.lean`, `LagrangianScreening.lean`, `StiefelHolonomy.lean`, ...) are recorded as named roots whose standing is certificates of form, pending the rebuild.

Leibniz receipts. Standing rule: primary sources only (*drive-triage/LEIBNIZ.md* locators), analogy-only unless the map is exhibited. This paper exhibits no Leibniz map and cites none.

7 Computational probes—toy models

The five witness scripts in `zenodo-audit/witness/` are carried here **rewritten as toy models**. Every "VERIFIED" line is dropped. None of them proves anything about the Clay Mathematics problems they gesture at; per RY's standing order the millennium attack is deferred to last, and the convergence target for the Clay work stays unnamed until it is unparked. Terminology rule observed throughout: "Clay Mathematics problems" / "CMPs".

- **Yang–Mills probe:** a 4D SU(2) lattice with plaquette Wilson loops; a toy illustration of gauge-link holonomy, not a mass-gap argument.
- **Navier–Stokes probe:** a 3D Burgers proxy for the incompressible equations with a chiral-gate illustration on enstrophy; a regularity toy, not an existence proof.
- **Riemann probe:** a zeta symmetry-error scan $|\zeta(s) - \overline{\zeta(1-\bar{s})}|$; a numerical illustration of critical-line symmetry, not a zero-location argument.
- **P vs NP probe:** a search-vs.-verify toy on random vectors against a target basis; an illustration of the asymmetry idiom, not a complexity result.
- **Hodge probe:** a cycle-alignment toy measuring projection of random classes onto an algebraic basis; an illustration, not a rationality argument.
- **Gravity probe (physics, not Clay):** a rotation-profile toy $v^2 = GM/r \times \text{tension}(r)$ illustrating the qualitative shape the χ-field would imprint; not a fit, not evidence.

8 PENDING on Grok's Res Nova stream

The following are **not claimed** and are not filled by invention. They are the flagged inputs to the Res Nova rebuild stream (information-tension theory first, per the standing plan), which feed this flagship as they land:

1. The derivation of information-tension theory proper: what χ *is*, from what it is derived, and why the AQUAL kinetic functional is the one it takes.
2. The migrated Lagrangian on the live substrate (the source's retired-frame banner, §3).

3. The spectral-action question: exhibit the map from the seven-sector sum to a spectral action, or prove the non-isomorphism (§5a).
4. The $E_8 \times E_8$ landing of the gauge sector (§5c).
5. Precision anchors mapped to this paper by triage but not available in text at draft time: folio 12 (electron mass from E_8 harmonics), folio 15 (muon-trap δ shift), folio 28 (Penning-trap inference). Not incorporated; not asserted.

9 What was left out and why

- The monograph's "zero free parameters" / "closed, parameter-free" claims: dropped—the sectors above carry explicit couplings (α, β, γ_0, $Y_{ff'}$, λ_ϕ).
- "Lossless inverse-ZIP decompression of an E_8 holonomy manifold": dropped—no map exhibited; analogy at best, and the isomorphism rule forbids the bare claim.
- $\Omega_\Lambda = \ln 2$, DESI concordance numbers, the S8 "resolution", SPARC 99.94% compliance, the Kerr master invariant, $g_{\text{weak}} = \kappa\chi/\pi$ numerics: dropped—asserted without kill criteria; empirical standing belongs to the sibling lanes with their criteria, not to this action paper.
- ADCCL / sovereign cognition: belongs to **the Observer** angle (successor (F)), not to this action paper.
- TTEY drive engineering and patent material: separate engineering track per triage (IP boundary is RY's call).
- Clay-problem proof claims in the witness README ("formal solutions to five of the six remaining . . . problems"): withdrawn; the probes are toy models (§7).
- The old record's "Machine-Verified (Lean 4: 0 Errors, 0 Warnings)" badge: withdrawn per the certificate rule (§6).